Problem-Solution Fit

Date	07 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Retail banks/NBFC & fintec underwriting team that process credit-card applications, and the individual applicants themselves including new-to-credit / thin-file applicants.	**6. CUSTOMER LIMITATIONS [CL]** Little or no bureau history for new-to-credit applicants; incomplete form fields; very few confirmed "bad" outcomes to learn from (severe class imbalance); limited compute budget for real-time scoring privacy / regulatory limits on which attributes can be used.	**5. AVAILABLE SOLUTIONS** PROS & CONS**[AS]** Manual review (handles nuance, but slow & inconsistent); static rule-based scorecards (fast & explainable, but rigid and need constant re-tuning); generic bureau credit scores (industry standard, but unusable for thin-file applicants).
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** Manual / rule-based review is slow (often days) and inconsistent across reviewers. This bottleneck recurs on every single application -a daily, high-volume pain point.	**9. PROBLEM ROOT / CAUSE [RC]** Risk labels were assigned from "ever had a serious delinquency," regardless of how long account was actually observed. Vintage analysis show cumulative bad-rate curves are still rising at 60 months on book, so short-history applicants get mislabeled "good" simply because they haven't been tracked long enough.	**7. BEHAVIOR** + ITS INTENSITY **[BE]** Applicants expect a near-instant response the moment they submit — intensity is high, especially on digital channels. Underwriters, by contrast, process applications in steadier batches/queues.

3. TRIGGERS TO ACT [TR]

Month-end application backlogs; a new-to-credit applicant with a thin file; a digital-first lender needing an instant online decision instead of manual queueing.

4. EMOTIONS BEFORE / AFTER **[EM]**

Before: applicant anxiety from an uncertain multi-day wait; underwriter fatigue from repetitive manual review.

After: applicant gets a quick, clear answer; underwriter trusts a consistent second opinion and can focus effort only on borderline cases.

10. YOUR SOLUTION [SL]

An XGBoost classification model trained on vintage filtered labels (only applicants with ≥24 months of observed credit history are labeled, removing label noise from prematurely-judged accounts), using encoded demographic, income, and employment features. The model is tuned via cross-validate hyperparameter search and an F1-optimized decision threshold (0.29), then served through a lightweight Flask web form that returns an Approve/Reject decision with its probability — giving underwriters a fast, consistent, and more honestly-labeled second opinion, while disclosing the model's real precision/recall instead of overstating reliability.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Digital application portals, mobile banking apps, fintech aggregators.

OFFLINE

Branch walk-ins, paper application forms, phone-based applications